

BUILDING STOCK ENERGY WORKBENCH

Valencia Simulation Report

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Run: ALL_VALENC-A_REAL

This report documents the annual Valencia building-stock simulation executed in Building Stock Energy Workbench (BSEW). It is designed to answer four questions: what evidence entered the run, which modelling and execution parameters were applied, which quality gates controlled acceptance, and what the settled results support academically.

SETTLED EVIDENCE — COVERAGE QUALIFIED. Process absence, final aggregate settlement, single-identity reconciliation, export integrity and document review passed. Results describe the represented successful stock; failed and excluded references remain explicit in coverage.

Suggested citation

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ALL_VALENC-A_REAL. 28 August 2026. Software commit
0836bdb1c689a34d13d1af76f764ed21f42a112b.

1. Executive interpretation

ALL_VALENC-A_REAL is a full-scope annual simulation of the supplied Valencia building stock under one recorded verified model profile. BSEW converts cadastral and dwelling evidence into simplified OpenStudio models, executes EnergyPlus, records terminal evidence per building and derives aggregate energy, operational-carbon, coverage, geometry-quality and QA summaries.

This edition publishes final values after all latest ledger records reconciled with the settled scope. The principal result classes will be total site energy, whole-site energy intensity under explicitly named residential-area denominators, end-use energy, operational carbon, cluster/district summaries, coverage, and diagnostic status.

The results are simulations under recorded assumptions. They are not measured utility consumption, per-building calibration, an energy certificate, regulatory compliance, embodied carbon or a guaranteed forecast of future operation. The verified profile establishes method consistency against the project's reference evidence; it does not establish measurement validation for every stock building.

1.1 Publication decision

This edition assigns **PUBLISHABLE WITH EXPLICIT COVERAGE LIMITS** after reconciling successful, failed, QA-rejected and excluded records. Building-level records remain individually classifiable as PASS, ACCEPTED WITH CLASSIFIED DIAGNOSTICS, FAILED / DO NOT USE, NOT ASSESSED or INCOMPLETE EVIDENCE — DO NOT CITE.

1.2 Final evidence gate

Table 1. Final evidence gate		
Gate	Required final evidence	Final state
Process	Run reports false and recorded PID is absent	PASS
Scope	Final terminal records reconcile to recorded scope	PASS
Aggregate	Final aggregate exists, is readable and is not partial	PASS
Identity	Latest accepted rows share intended run/profile/input identity	PASS
Exports	CSV, GIS, heat map and manifest reconcile to ledger	PASS
Review	QA, diagnostics, limitations and selected buildings reviewed	PASS

2. Research objective and analytical boundary

The simulation provides reproducible evidence for the spatial and typological distribution of modelled final site energy across the supplied Valencia stock. It supports comparison among recorded building

clusters and declared area bases, inspection of preparation and EnergyPlus evidence at building level, and production of traceable row-level and spatial exports.

The analytical unit is the building or cadastral reference represented by the latest terminal ledger record. Aggregate values include only records admitted by the stated result and QA contract. Excluded and failed records remain visible in coverage evidence and are not interpreted as zero energy.

2.1 Questions the report can answer

- What final site energy and operational emissions were simulated for the represented stock?
- How do results distribute across building clusters and, where supported, districts?
- How do geometric and cadastral residential-area denominators change intensity interpretation?
- Which buildings failed or were excluded, and what bias can or cannot be quantified?
- Which preparation, geometry, occupancy and QA assumptions constrain interpretation?

2.2 Questions it cannot answer alone

- Whether a specific building matches its utility bills or actual occupancy.
- Whether the stock is calibrated against measured consumption.
- Whether a simulated difference is causally attributable to an intervention.
- Whether operational factors represent embodied or whole-life carbon.
- Whether simplified one-zone-per-storey geometry captures all local thermal diversity.

3. Run identity and execution record

This edition reads identity from the settled run directory and reconciled ledger/configuration evidence, not from mutable Files-page state. The run name is a human-recognisable label; cryptographic fingerprints establish the resolved scientific identity.

Table 2. Run identity and execution record		
Identity element	Authoritative source	Final value
Run name	Run directory and API route	ALL_VALENC-A_REAL
Run identity	Configuration / latest ledger	8def48ce840d8a1e... (full value in evidence manifest)
Verified profile	Configuration / latest ledger	cbf17543cfee3ff1... (full value in evidence manifest)
Climate fingerprint	Configuration / latest ledger	79c11a5992edeee5... (full value in evidence manifest)
Template fingerprint	Configuration / latest ledger	a5d08275f02c9d76... (full value in evidence manifest)
Policy fingerprint	Configuration / latest ledger	6b730d4946e772c1... (full value in evidence manifest)
Stock-source fingerprint	Configuration / latest ledger	05736c0028799d84... (full value in evidence manifest)
Runner schema	Configuration / latest ledger	2
Execution segments	Preserved process records	segment 1: 8 workers, full retention, resume

The production interface submits six workers and full per-building retention for a new UI request; these are not user-editable controls on the Run page. This run has an interruption/resume history, so execution provenance is reported by preserved segment rather than inferred from the UI default. Worker count affects throughput rather than frozen model inputs. Timing claims are limited to preserved process and ledger evidence.

3.1 Period

The Valencia run is annual. Final energy is reported as kWh/year or GWh/year; intensity as kWh/m²·year; operational carbon as kgCO₂/m²·year or tCO₂/year where the corresponding denominator is explicit. No microclimate event values will be mixed into this annual ledger.

4. Authoritative inputs

The run resolves its active input set once and records content identities. Absolute workstation paths are excluded from this report. Logical dataset role, preserved filename, validation contract and SHA-256 evidence are retained.

4.1 Valencia cadastre

The building GIS supplies parcel/building geometry and attributes used for identity, scope, storeys, population, dwellings, cluster, district/context and footprint preparation. It is an administrative/research source, not a geometric survey. The evidence manifest records the managed filename, component hashes, prepared-stock identity and field-role policy; spatial export acceptance records CRS and feature reconciliation.

4.2 Tipo15 dwelling ledger

The companion semicolon-delimited Latin-1 CSV supplies 31_pc, 252_planta, 428_uso and 442_sup_Residencial. It supports parcel join, floor/use interpretation and the cadastral residential-area denominator. Multiple dwelling records per parcel are expected. Missing joins and fallback use are quantified rather than converted to zero.

4.3 EPW and DDY climate pair

The annual EPW supplies hourly weather and site metadata; the matched DDY supplies winter and summer design days. The pair is validated and activated atomically. Slab-contact ground temperature and water-mains temperature are recorded separately because an EPW does not authoritatively provide both model inputs.

4.4 OpenStudio template

The OSM template provides role-bound constructions, schedules, thermostats, space types and required system objects. Its fingerprint identifies the source library used to build every accepted model; it is not a measured representation of each building.

Table 3. OpenStudio template			
Dataset role	Preserved filename	Contract	SHA-256 / fingerprint
Building GIS	DatosRai_ciudadValencia.shp	Shapefile collection + active policy	5 component SHA-256 values in manifest; 05736c0028799d8 (full value in evidence manifest)
Tipo15	2_daire_kutugu_Tipo15_soloV(in).csv	tipo15-v1	c6e85c677c32596 (full value in

Dataset role	Preserved filename	Contract	SHA-256 / fingerprint
			evidence manifest)
Annual EPW	3_iklim_ESP_Valencia.082840_IWEC.epw	epw- annual-v1	09dd4adf2bf25ba5 (full value in evidence manifest)
DDY	4_tasarim_gunleri_ESP_Valencia.082840_IWEC.ddy	ddy- design- days-v1	3ddb6f02ca132080 (full value in evidence manifest)
OpenStudio template	PlantillaOS_v2.osm	template- roles-v1	292dcf0fcecac1b98 (full value in evidence manifest)

4.5 Extracted data

This section states exactly which data were taken out of the supplied sources and which were produced by the run. A field that exists in a source file is not evidence that the model consumed it.

4.5.1 Fields extracted from the inputs

The Valencia cadastre carries 35 attribute fields plus polygon geometry. The preparation policy carries **seven** of them into the engine-ready stock; the remainder are not read during simulation.

Table 4. Fields extracted from the inputs		
Source	Extracted and used	Role in the model
Cadastre	<code>refparcela</code>	Building identity, ledger key, run scope
Cadastre	<code>altura_max</code>	Recorded storeys, before the cadastre storey rule
Cadastre	<code>cluster</code>	TABULA family/period, from which envelope U-values resolve
Cadastre	<code>nombre</code>	District grouping for aggregate reporting
Cadastre	<code>pob_total</code>	Resident population driving occupancy and hot-water draw
Cadastre	<code>num_vivend</code>	Dwelling count for per-dwelling intensities
Cadastre	<code>polygon geometry</code>	Footprint preparation, adjacency, party surfaces, context shading

Source	Extracted and used	Role in the model
Tipo15	31_pc	Join key from dwelling record to cadastral parcel
Tipo15	252_planta	Floor label used for ground-use interpretation
Tipo15	428_uso	Use code separating residential from other floors
Tipo15	442_sup_Residencial	Positive residential area forming the cadastral denominator
EPW	8,760 hourly weather rows and the site record	Annual boundary conditions and site identity
DDY	Winter and summer design days	Equipment sizing conditions
Template	Role-bound constructions, schedules, thermostats and space types	Operating regime applied to every building

The prepared stock written from these sources holds sixteen columns: the seven cadastral fields above plus nine derived ones (`family`, `period`, `footprint_area_m2`, `imputed_floors`, `res_area_proxy`, `dup_refparcela`, `ground_use`, `ground_use_source`, `tipo15_res_area_m2`). Derived fields are transparent preparation results, not new measurements.

Fields deliberately not extracted. `coorx`, `coory`, `pob_0_14`, `pob_15_65`, `pob_66_mas`, `the_geom_L`, `the_geom_A`, `Shape_Leng`, `Shape_Area`, `referencia`, `codigo_ine`, `nombre_mun`, `codigo_pro`, `zona_clima`, `uso_princi`, `altura_m_1`, `tipologia_`, `ano_constr`, `ano_cons_1`, `numero_viv`, `coddistrit`, `demanda_ca`, `demanda__1`, `calificaci`, `califica_1`, `coste_inte`, `coste_in_1`, `ConsumE` and `ConsumETot` are present in the source but are not read by the simulation. Three of these deserve an explicit note:

- `ano_constr` is not read directly because the construction period already reaches the model through `cluster`.
- `numero_viv` and `num_vivend` are two dwelling counts in the same file; only `num_vivend` is used, and the choice is recorded rather than inferred.
- District grouping is keyed on `nombre` rather than `coddistrit` because the two disagree for at least one record in this source; the name is the field the aggregate uses.

`demanda_ca` (an external certificate heating demand) and `ConsumE` (a per-cluster reference consumption) are **reference material, not model input**. The Rai comparison column shown in the interface is resolved from the published per-cluster reference table, not by reading the shapefile column at run time.

The Tipo15 ledger carries seventeen columns; the four listed above are read and the remaining thirteen (`29_cn`, `126_nm`, `201_tv`, `206_nv`, `231_numPolicia`, `246_bloque`, `250_escalera`, `255_puerta`, `283_codigoPostal`, `288_distrito`, `368_numOrden`, `372_ant`, `452_supSolar`) are not.

4.5.2 Fields produced by the run

The run writes one durable ledger record per building and one final aggregate. The curated Building CSV publishes **88 fields per building**, grouped as follows.

Table 5. Fields produced by the run		
Group	Fields	Examples
Identity and status	5	refparcela, status, reason, message, cluster
Period and event	5	run_mode, event_days, event_window, delta_peak_k, delta_base_k
Execution	2	seconds, pruned_bytes
Energy	24	total_site_kwh, total_site_kwh_m2, space_heating_kwh_m2, cooling_kwh_m2, dhw_kwh_m2, site_elec_kwh_m2
Operational carbon	4	total_site_co2_kg_m2, hvac_co2_kg_m2, total_site_co2_t, hvac_co2_
Geometry and area	24	footprint_m2, res_area_m2, tipol5_res_area_m2, built_storeys, top_storey_fraction, footprint_fidelity, n_windows
Occupancy and use	9	pob_total, num_vivend, occupants_applied, occupants_source, occupancy_plausibility, ground_use
Construction	2	wall_construction, roof_construction
Quality assurance	6	qa_all_passed, warnings, severes, severes_benign_shading_ems, severes_unexplained, fatals
Provenance	7	profile_fingerprint, climate_fingerprint, template_fingerprint, policy_fingerprint, stock_source_fingerprint, runner_schema, zero_policy

The exported data dictionary names every published field beside its source, and the User Guide reproduces the same ordered list.

The final aggregate carries **21 blocks**: totals, by_cluster, by_district, coverage, energy_period, buildings_ok, buildings_failed, buildings_failed_qa, buildings_excluded, qa_failed, unexplained_severes, implausible_occupancy, geometry_quality, floor_area_allocation, fragmentation, zoning, results_layer, ledger, provenance, elapsed_minutes and seconds_per_building. Absence of a block means the quantity was not measured for this run; it never means zero.

5. Software and verified profile

The documented toolchain is Python 3.13, OpenStudio 3.11.0 and EnergyPlus 25.2.0 [1]–[3]. The publication records software commit `0836bdb1c689a34d13d1af76f764ed21f42a112b`; release/toolchain verification retains the available patch-level version evidence. Dependency presence is a reproducibility condition, not evidence of physical accuracy.

The verified profile pins the model-building sources, template identity, policy and acceptance evidence. Profile verification means that the modelling method passed the project's fixed reference gates. It does not mean that Valencia's individual buildings were calibrated to measurements.

5.1 Profile-integrity rule

Every accepted ledger row must carry the intended profile fingerprint. Source drift during execution is not silently accepted. A mechanically combined ledger containing more than one profile cannot be published as one homogeneous run without explicit separation and warning.

5.2 Frozen and derived evidence

Physical model outputs are produced by the pinned simulation chain. Reporting may format units and labels, select latest records, create deterministic aggregates and check duplicate evidence for conflicts. It must not silently recompute historical physical results with today's changed model rules.

6. Building preparation and geometry

The pipeline transforms supplied polygons into tractable building masses, identifies context and party surfaces, applies storey rules, assigns ground use, creates openings where exterior walls exist, and records fidelity evidence.

6.1 Footprint preparation

Simplification fidelity is measured as symmetric difference divided by raw area. The configured bound and selected tolerance are recorded per building. A refined tolerance or as-drawn fallback protects geometry when the default tolerance would exceed the fidelity bound. A high fidelity score means closeness to the supplied polygon under this transformation; it does not prove that the source polygon is survey-accurate.

6.2 Storeys and top-storey allocation

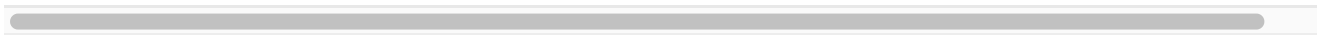
The report distinguishes total levels, residential levels above ground, effective residential storeys, built storeys and top-storey fraction. A recorded storey snap may remove a resolution-scale sliver near an integer boundary. Dwelling loads on a partial top storey are scaled according to the recorded fraction; the physical model still uses whole geometric storeys.

6.3 Zoning and context

The production scheme uses one well-mixed thermal zone per storey. Large buildings above the recorded threshold do not receive a core/perimeter subdivision. Party surfaces, context shading and openings are counted per accepted model. A ground storey with no exterior wall can legitimately have zero glazed subsurfaces.

Table 6. Zoning and context	
Geometry evidence	Final statistic
Buildings with measured fidelity	26,406
Median / P95 / maximum fidelity	0.000587 / 0.002898 / 0.009982
Configured / refined / coarser / as-drawn	26,327 / 44 / 35 / 0
Storey-snapped buildings and area share	579; 3.19%
Large single-zone buildings and energy share	122; 5.76%

Geometry evidence	Final statistic
Party surfaces — median / P95 / maximum	10.00 / 35.00 / 160.00 (26,406 measured)
Context shading surfaces — median / P95 / maximum	82.00 / 204.00 / 437.00 (26,406 measured)
Windows — median / P95 / maximum	18.00 / 138.00 / 1,506.00 (26,406 measured)



7. Use, occupancy and domestic hot water

Ground-storey use is conditional on the final recorded building evidence. Residential ground levels are treated as dwelling space and included in the residential basis. Commercial ground levels follow the verified tertiary regime and are excluded from a residential-only denominator. Mixed or unresolved evidence is not forced into either narrative.

Occupancy reporting separates administrative Padrón population, occupants applied to the model, source/fallback, density cap and plausibility status. Padrón is not live presence. A measured zero, an imputed value and a capped value are different evidence states.

Domestic hot water follows the recorded occupancy and model policy. The cited CTE reference demand of 28 litres/person/day is a 60 °C reference [6]; the model's 50 °C implementation is not claimed to be directly equivalent. This edition states both temperatures and the modelling interpretation.

7.1 Mixed-use energy attribution

Commercial lighting and equipment can be separated by user-defined subcategory. HVAC and DHW are not fully separable by space type under the preserved meter contract. Therefore `residential_site_kwh` is a **residential-attributed upper bound**, not exact residential consumption.

8. HVAC, schedules and climate application

The generated models use the constructions, thermostat schedules, internal-load schedules, space types and HVAC objects bound in the verified template. The methods table identifies the relevant evidence families and relies on the preserved model/template records rather than retyping unsupported assumptions.

Annual weather is read from the recorded Valencia EPW. Design-day objects are selected from the matched DDY. Barometric pressure and site metadata are derived from the active climate contract. The annual Valencia report does not include or annualise the Lecco PALM microclimate slice; that evidence belongs to the separate `LECCO_1` event run.

8.1 Parameters to publish

Table 7. Parameters to publish	
Parameter family	Publication evidence
Heating/cooling thermostat regimes	Template roles and generated OSM
Infiltration and ventilation assumptions	Verified profile/template evidence
Lighting/equipment schedules	Template roles and subcategory evidence
HVAC system and fuel	Generated model and EnergyPlus tables
DHW demand and temperature	Model preparation record
EPW/DDY site and design days	Climate manifest
Ground and water-mains temperatures	Run climate record

9. Result and denominator contract

Every final metric is presented with visible name, authoritative field, unit, period, denominator, precision and interpretation limit. Missing optional evidence is an em dash; a measured zero remains zero. Boolean, NaN, infinity and wrong-type values are never formatted as valid measurements.

9.1 Principal energy quantities

Total site energy is the sum of simulated final energy admitted by the result contract. End-use reporting includes space heating, cooling, domestic hot water, fans, pumps, lighting and equipment where recorded. EnergyPlus ABUPS values originate in output meters; user-defined subcategories provide applicable splits [4].

9.2 Area denominators

- **Geometric residential area:** modelled residential-storey basis used by the comparison convention.
- **Cadastral residential area:** Tipo15 administrative area where available.
- **Conditioned area:** all conditioned model area, including applicable non-residential space.
- **Residential-attributed upper bound:** energy remaining after separable commercial categories are removed; not exact attribution.

Whole-site energy divided by residential-only area is a comparison convention. It can include commercial ground-floor energy in the numerator while excluding that floor from the denominator. This is not the physical intensity of all conditioned space.

10. Operational carbon

Operational emissions are derived from simulated final energy using the recorded project factors: 0.331 kgCO₂/kWh_final for electricity and 0.252 kgCO₂/kWh_final for natural gas [5]. The table below reconciles settled aggregate carbon with named presentation denominators.

The result excludes embodied carbon and is not a measured emissions inventory. Factors may not represent another reporting year, electricity supplier or marginal-emissions question. Carbon figures remain tied to the annual period and the stated energy basis.

Table 8. Operational carbon			
Carbon quantity	Unit	Denominator	Final value
Total operational carbon	tCO ₂ /year	None	661,312.30
Operational carbon intensity	kgCO ₂ /m ² ·year	Geometric residential area	13.89
Operational carbon intensity	kgCO ₂ /m ² ·year	Conditioned area	11.73
HVAC operational carbon	tCO ₂ /year	None	76,279.90

11. Coverage, failures and exclusions

Coverage is reported as successful result count divided by buildings in scope, together with failed, QA-rejected and excluded counts. A high percentage does not prove absence of bias. Missing buildings may differ systematically in size, cluster, geometry or use.

When a common cadastral-area basis exists for represented and missing records, the report may size potential truncation under the recorded convention. Otherwise the correct statement is that truncation is unquantified; it is never “no bias.”

Table 9. Coverage, failures and exclusions	
Scope accounting	Final count
Buildings in scope	26,445
Runnable after preflight	26,407
Successful latest records	26,406
Runtime failed	1
QA rejected	0
Pre-simulation excluded	38
Coverage percentage	99.85%

Failure and exclusion reasons are grouped without erasing their references. Repeated attempts are resolved by latest-record semantics; attempts are not double-counted as buildings.

12. Quality assurance and diagnostics

The building QA matrix compares expected model evidence, observed EnergyPlus evidence or accepted method interval, criterion and result. Stored ratio tolerances are reported in reader-facing units: 0.005 as $\pm 0.5\%$, 0.02 as $\pm 2\%$, and annual unmet hours as ≤ 500 h.

Some checks compare the built model with EnergyPlus output. A plausibility-band check evaluates the method against a research range and is not mislabelled as an EnergyPlus cross-check.

12.1 Diagnostic classification

EnergyPlus Severe messages generally require correction [7]. BSEW can accept only a narrowly recorded project-classified benign pattern when all QA checks pass and no Fatal or unexplained Severe remains. Such a building is **ACCEPTED WITH CLASSIFIED DIAGNOSTICS**, not PASS. This edition discloses classified-message counts and the project-specific acceptance rule; row-level markers remain in preserved evidence.

Table 10. Diagnostic classification	
QA/diagnostic evidence	Final result
QA rejected buildings	0
Accepted rows with classified benign Severe	26,294
Project-classified benign Severe messages	1,042,465
Unexplained Severe messages	0
Fatal messages	0
Occupancy plausibility flags	3,954

13. Aggregate results and cluster interpretation

The results table reports energy and operational carbon from settled evidence. Cluster rows include building count, residential area, total site energy and the available geometric/cadastral intensities. Rai comparison columns are shown only where the published comparison exists and the denominator is aligned.

Cluster intensity is energy divided by the corresponding total area. It is not the arithmetic mean of building intensity. A difference from the Rai reference is a method/reference comparison, not calibration against utility observations.

Table 11. Aggregate results and cluster interpretation		
Result	Unit and basis	Final value
Total site energy	GWh/year, represented stock	2,084.037
Space heating	GWh/year	122.330
Cooling	GWh/year	91.624
Domestic hot water	GWh/year	360.792
Whole-site / geometric residential area	kWh/m ² ·year	43.77
Whole-site / cadastral residential area	kWh/m ² ·year	47.08
Whole-site / conditioned area	kWh/m ² ·year	36.95
Operational carbon	tCO ₂ /year	661,312.30

13.1 Cluster results

Table 12. Cluster results					
Cluster	Buildings	Residential area [m ²]	Site energy [GWh/year]	Geometric [kWh/m ² ·year]	Cadastral [kWh/m ² ·year]
BlocPluriP04	8,092	20,461,540.5	933.805	45.64	48.16
BlocPluriP05	3,759	14,125,743.2	593.298	42.00	43.91
BlocPluriP03	3,124	5,023,841.7	230.041	45.79	49.59
BlocPluriP02	2,048	2,479,648.1	109.694	44.24	49.49
BlocPluriP06	537	1,729,270.3	62.551	36.17	39.89
EdiPluriP05	363	631,992.4	25.817	40.85	50.02
BlocPluriP01	664	457,992.9	19.990	43.65	49.09
EdiPluriP02	1,404	427,119.7	18.124	42.43	52.15
VivUniP02	1,697	392,508.8	17.133	43.65	65.95
VivUniP04	1,085	358,854.7	14.854	41.39	70.22
VivUniP05	940	383,970.1	13.684	35.64	48.61
VivUniP03	949	255,594.2	10.756	42.08	61.71
EdiPluriP03	656	241,656.7	10.314	42.68	53.15
VivUniP06	373	225,566.6	6.971	30.91	36.53
EdiPluriP04	298	136,012.0	6.624	48.70	54.81
VivUniP01	183	65,279.6	2.774	42.50	43.91
BlocPluriP07	14	86,453.9	2.736	31.64	37.02
EdiPluriP01	139	55,018.7	2.450	44.53	59.06
EdiPluriP06	71	65,882.4	2.300	34.91	42.48
VivUniP07	9	5,108.3	0.112	21.87	43.95

Cluster	Buildings	Residential area [m²]	Site energy [GWh/year]	Geometric [kWh/m²·year]	Cadastral [kWh/m²·year]
EdiPluriP07	1	351.0	0.009	25.80	35.10

13.2 District results

Table 13. District results				
District	Buildings	Residential area [m²]	Site energy [GWh/year]	Geometric [kWh/m²·year]
QUATRE CARRERES	1,699	3,797,972.0	171.352	45.12
CAMINS AL GRAU	1,288	3,637,549.5	157.793	43.38
L'EIXAMPLE	1,969	3,464,625.2	148.206	42.78
EXTRAMURS	1,844	3,449,296.1	148.200	42.97
POBLATS MARITIMS	4,057	3,120,382.5	140.266	44.95
PATRAIX	1,014	3,125,435.7	137.320	43.94
JESUS	1,132	2,640,860.9	118.673	44.94
RASCANYA	1,166	2,548,294.7	115.772	45.43
LA SAIDIA	1,225	2,549,779.0	114.220	44.80
L'OLIVERETA	1,238	2,364,385.2	109.391	46.27
CIUTAT VELLA	2,329	2,561,117.6	107.964	42.16
BENICALAP	1,009	2,354,263.0	103.998	44.17
ALGIROS	630	2,246,242.0	97.308	43.32
CAMPANAR	688	2,246,628.5	96.090	42.77
EL PLA DEL REAL	539	2,176,879.5	90.933	41.77
POBLATS DEL SUD	1,674	2,002,802.3	83.794	41.84
BENIMACLET	950	1,728,044.2	74.437	43.08

District	Buildings	Residential area [m²]	Site energy [GWh/year]	Geometric [kWh/m²·year]
POBLATS DE L'OEST	1,164	955,900.8	41.509	43.42
POBLATS DEL NORD	791	638,947.1	26.810	41.96

14. Spatial evidence and exports

The final GeoPackage passed the accepted export checks for CRS, feature count, one feature per latest building reference, null-energy representation of failed/excluded records, context fields, derived per-person/per-dwelling fields, cluster/district layers and supplied QGIS style [8].

The heat map is a publication-oriented image. Its caption identifies run, annual period, metric, denominator, coverage, colour-scale clipping and the simulated-not-measured boundary. Buildings without results will remain visible in a separate status treatment. Distant valid geometries will not be silently cropped from every view.

The Building CSV is the authoritative row-level analysis export after schema correction. Final acceptance compared its headers and latest-record contents with the grouped User Guide field appendix. Absolute local paths will not appear in the curated publication schema.

14.1 Export integrity

Every accepted DOCX/PDF/HTML report, CSV, GeoPackage, heat map and evidence manifest has a recorded SHA-256 in the publication manifest. Signed packages contain an Ed25519 manifest. Hash/signature verification establishes integrity, not scientific validity.

14.2 Accepted artifact hashes

Table 14. Accepted artifact hashes	
Accepted artifact	SHA-256
aggregate.json	d4dee63a1e10f2ce16c8ac57b174f21eecd91a41abfa20fa1eb42bdf0c585f89
buildings.csv	9ae87afcfc41bda199002cf55d708c9450dc37b2df318c624a55d536cf9f5897
buildings_dictionary.csv	3b284be41bba7615de38e8a910c79ef0f61877d667820ed9c9200327ec5e2778
ledger.jsonl	244e0f1ff407e54f3edaab347537ce60a3fdb9d43d496dea396dea547c876338
results_buildings.gpkg	34c235c8b04ac648fb75a9aeeb955cb3d3ffa44d871e6a51e455085a36f43da2
results_buildings.qml	1410137416eac5024aac6e27c583a4b6b1eb2f7d4badba51ad31ed0c0268c422
results_heatmap.json	99ba02a400aa10c113f323afaa7b0db0144164cb048169dec3323373d938ced5
results_heatmap.png	9c92626a8868be3b4c34b0348507c5299f3b68cd0a68b6c2b4615bbf4fb39e53
run_config.json	30f5da100bd69ab03f821f94241c6b6ebd38923c4b4973228355d0f66c5e7f61
run_process.json	4b13c6610bc97d2b157646b876692f1c6d46ace922a0866b000c9c26bc8bcf63

Accepted artifact	SHA-256
valencia-publication-evidence.json	6bfc1cf434e591ccc194be3fc4c8b042ffe3bcb2c5798b6d3af1ea3b58b12793



15. Reproducible representative-building selection

This edition selects up to six references by deterministic rules applied to settled latest records. Duplicate selections are removed in the order below and the exact algorithm, source hash and selected references are preserved in the evidence manifest.

- 1. Successful building nearest the median valid total-site intensity.
- 2. Successful building nearest the fifth percentile.
- 3. Successful building nearest the ninety-fifth percentile.
- 4. Successful building with the greatest total-site energy contribution.
- 5. Accepted building carrying a recorded occupancy or classified-diagnostic flag, if available.
- 6. One failed or excluded reference representing the most frequent non-success reason, if available.

For successful examples, the evidence manifest preserves identity, cluster, area bases, energy, carbon, occupancy, geometry, QA and diagnostic fields; the table below provides the citation-oriented summary. The failed/excluded example will not be assigned energy; it will document the failure stage and reason.

Table 15. Reproducible representative-building selection		
Selection role	Reference	Evidence status
median-nearest	5327403YJ2752G	ACCEPTED WITH CLASSIFIED DIAGNOSTICS
p5-nearest	9504916YJ2790D	ACCEPTED WITH CLASSIFIED DIAGNOSTICS
p95-nearest	9826309YJ2792F	ACCEPTED WITH CLASSIFIED DIAGNOSTICS
highest-total-contribution	4889401YJ2648H	ACCEPTED WITH CLASSIFIED DIAGNOSTICS
flagged-accepted	0022301YJ3702C	ACCEPTED WITH CLASSIFIED DIAGNOSTICS
failed-or-excluded	0023911YJ3702C	EXCLUDED — footprint_outside_range

16. Limitations and responsible use

The final interpretation retains the following limits beside affected findings:

- cadastral geometry, use and area are administrative evidence with measurement/allocation limits;
- one parcel is not always one physical building;
- simplified zoning and integer-storey geometry affect area representation;
- large one-zone-per-storey buildings do not represent core/perimeter diversity;
- difficult/courtyard geometry can be excluded by preparation gates;
- occupancy uses administrative evidence, caps and recorded imputation;
- HVAC and DHW are not exactly separable between residential and commercial spaces;
- operational carbon excludes embodied carbon;
- profile verification is not per-building measurement calibration;
- pilot-building LHS evidence is not a city-total uncertainty band;
- failed/excluded records can create unquantified truncation.

These limitations do not automatically invalidate the study. They define the claims the evidence can support and the additional measurements or sensitivity analyses needed for stronger claims.

17. Provenance, reproducibility and publication package

The final publication package preserves logical filenames and hashes for the run configuration, process record, ledger, final aggregate, corrected Building CSV, GeoPackage/style, heat map, selected readable reports, EnergyPlus tables, technical evidence and document outputs. Absolute workstation paths will be removed from publication-facing documents and curated exports.

No result will be accepted when duplicate authoritative evidence blocks conflict. A building identity mismatch prevents report generation. A successful-looking row without essential period, result, QA or diagnostic evidence is INCOMPLETE EVIDENCE — DO NOT CITE.

17.1 Evidence manifest

The machine-readable JSON/CSV manifest contains:

- run, profile, climate, template, policy, stock-source and runner-schema identities;
- final scope/count/period reconciliation;
- source and output SHA-256 values;
- key method parameters and units;
- selected-building algorithm and references;
- figure/table source files;
- software commit and publication date;
- review gates and their PASS/FAIL state.

18. References

[1] Python Software Foundation, “Python 3.13 documentation,” <https://docs.python.org/3.13/>

[2] NREL, “OpenStudio 3.11.0 release,” <https://github.com/NREL/OpenStudio/releases/tag/v3.11.0>

[3] EnergyPlus, “EnergyPlus 25.2 documentation,” <https://energyplus.readthedocs.io/en/v25.2.0/>

[4] U.S. Department of Energy, *EnergyPlus Input Output Reference*, version 25.2, <https://bigladdersoftware.com/epx/docs/25-2/input-output-reference/index.html>

[5] Instituto para la Diversificación y Ahorro de la Energía, *La bomba de calor en la rehabilitación energética de edificios*, 2023, https://www.idae.es/sites/default/files/documentos/publicaciones_idae/Guias_IDAE_La_Bomba_de_calor_2023_V11.pdf

[6] Gobierno de España, *Código Técnico de la Edificación, DB-HE*, 2017 edition, https://www.codigotecnico.org/pdf/Documentos/HE/DBHE_201706.pdf

[7] U.S. Department of Energy, *EnergyPlus Essentials*, version 25.2, <https://energyplus.readthedocs.io/en/v25.2.0/essentials/essentials.html>

[8] Open Geospatial Consortium, “GeoPackage Encoding Standard,” <https://www.ogc.org/standard/geopackage/>